

LOKESH PANDITI

Embedded Systems & Firmware Developer

lokesh.panditi.29@gmail.com | +91 9390978139 | [Linkedin](#) | [Github](#) | [Portfolio](#) | Ongole, India

Aspiring Embedded Systems and Firmware Developer with a strong interest in bare-metal programming, low-level software, and embedded systems. Skilled in Ada, C, C++, Python, Rust, and AArch64 Assembly, with hands-on experience building embedded frameworks and firmware for microcontrollers.

PROJECTS

Glyph — Embedded Graphics Framework

GitHub: <https://github.com/thirstymelon/glyph>

Ada • Embedded Systems • RP2040 • Graphics

- Developing a portable embedded graphics framework in Ada for resource-constrained microcontrollers.
- Provides a unified API for drawing graphics and text across different display hardware.
- Simplifies Embedded GUI development by abstracting display drivers and hardware interfaces.
- Initially targets RP2040 with support for additional displays and microcontroller platforms.

Radio_Kit — RF Signal Generation Framework

GitHub: https://github.com/thirstymelon/Radio_Kit

Ada • RP2040 • Bare-Metal

- Developing a bare-metal framework for generating radio-frequency signals using RP2040.
- Enables RF transmission without requiring dedicated RF transmitter hardware.
- Provides a reusable foundation for experimenting with embedded RF communication techniques.
- Designed to support multiple transmission methods and future RP2040-based platforms.

Web_Kit — Web Server Framework

GitHub: https://github.com/thirstymelon/Web_Kit

Ada • Embedded Systems • HTTP • RP2040

- Developing a lightweight Web framework in Ada for building web applications and REST APIs.
 - Provides reusable components for HTTP request handling, routing and middleware.
 - Simplifies backend application development with a modular and maintainable architecture.
 - Designed for building scalable server-side applications using Ada.
-

SKILLS

Programming Languages

Ada, C, C++, ARM Assembly (AArch64), Python, Rust (Beginner)

Embedded Platforms

RP2040, ESP32

Embedded Development

Bare-Metal Programming, Embedded Systems, Firmware Development, Driver Development

Communication Protocols

UART, SPI, I²C, CAN

Tools & Technologies

Git, GitHub, PlatformIO, ESP-IDF, VS Code

Core Concepts

Firmware Architecture, Hardware Abstraction Layers (HAL), Layered Software Design, Debugging

EDUCATION

Bachelor of Technology in Computer Science, Parul University

2023–2027